

6(a)	$I = I_1 + I_2 + I_3$	B1
	$(V/R) = (V/R_1) + (V/R_2) + (V/R_3)$ or $(I/V) = (I_1/V) + (I_2/V) + (I_3/V)$ <u>and</u> (so) $1/R = 1/R_1 + 1/R_2 + 1/R_3$	A1
6(b)(i)	e.m.f. is total energy available per unit charge	B1
	energy is dissipated in the internal resistance/resistor/ r	B1
6(b)(ii)1	Energy = EQ	C1
	$= 6.0 \times 2.5 \times 10^3$ $= 1.5 \times 10^4 \text{ J}$	A1
6(b)(ii)2	number = $2.5 \times 10^3 / 1.6 \times 10^{-19}$ $= 1.6 \times 10^{22}$ (1.56×10^{22})	A1
6(b)(iii)	$1/4.8 = 1/12 + 1/R_X$ $R_X = 8.0 \Omega$	A1
6(b)(iv)	$P = V^2/R$ or $P = VI$ <u>and</u> $V = IR$	C1
	ratio = $(V^2/8)/(V^2/12) = 12/8$ $= 1.5$	A1
6(b)(v)	(total) current, or I , increases <u>and</u> $P = EI$ or $P = 6I$ or $P \propto I$ or total (circuit) resistance decreases <u>and</u> $P = E^2/R$ or $P = 36/R$ or $P \propto 1/R$	B1